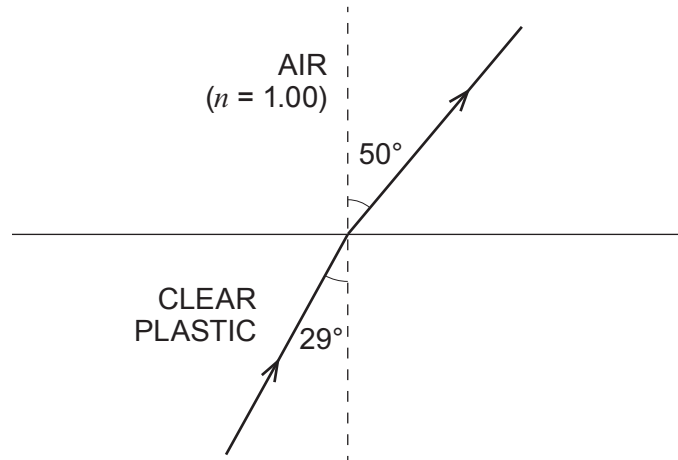


3. (a) A narrow beam of light is observed to refract as shown between clear plastic and air.



- (i) Calculate the speed of light in the **plastic**.

[2]

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- (ii) Calculate the critical angle for light approaching air from the plastic.

[1]

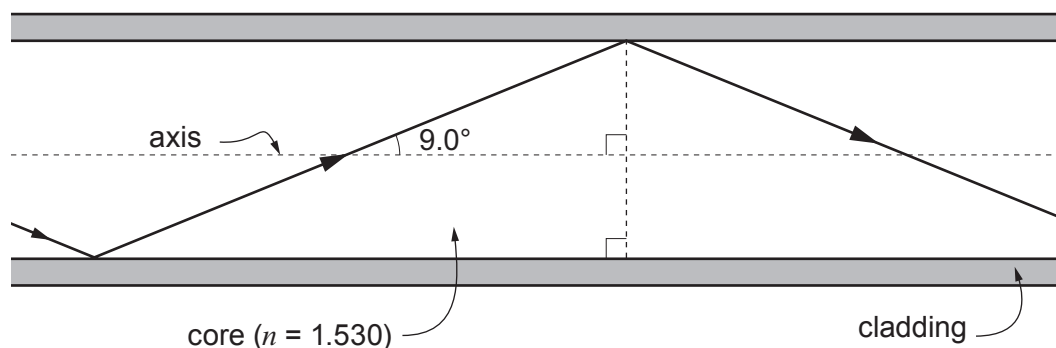
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- (b) The diagram shows light being transmitted along a multimode fibre, at the greatest angle to the axis for successful transmission.



NOT TO SCALE

- (i) The refractive index of the core is 1.530. Calculate the refractive index of the cladding. [2]

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- (ii) Show that the zigzag route in the diagram is 1.0125 times longer than a straight path through the same length of fibre. [1]

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- (iii) The difference in times of travel for a data pulse by these two extreme routes is required to be no more than 7.5 ns. Determine whether or not 150 m of fibre will be too long. Set out your reasoning clearly. [3]

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